

LEGAL AGCM, AB, Canada

WMO: 715440

Lat: 54.0031N

Lon: 113.4744W

Elev: 680

StdP: 93.42

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-30.8	-27.6	-34.2	0.2	-30.5	-30.8	0.2	-27.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	27.5	17.1	25.9	16.4	24.3	15.8	19.2	24.8	18.0	23.6	17.0	22.3	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.1	13.3	22.3	15.9	12.2	20.9	14.7	11.4	19.4	57.7	24.7	53.7	23.6	50.2	22.3	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
N/A	N/A	N/A	DB	-33.8	30.8	1.6	1.2	-34.9	31.6	-35.8	32.3	-36.7	32.9	-37.8	33.7
			WB	-33.9	21.7	1.5	1.4	-35.0	22.7	-35.9	23.5	-36.8	24.3	-37.9	25.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	3.0	-10.2	-10.8	-4.8	3.5	11.0	14.9	16.8	16.0	11.1	4.1	-4.5	-11.5	(6)
(7)		DBStd	12.09	8.75	8.34	7.73	5.15	4.68	2.97	2.55	3.21	4.51	4.88	6.91	7.92	(7)
(8)		HDD10.0	3251	625	582	461	202	44	3	0	2	40	191	436	666	(8)
(9)		HDD18.3	5618	883	816	719	446	229	109	61	85	219	441	686	924	(9)
(10)		CDD10.0	711	0	0	0	7	76	149	212	187	72	8	0	0	(10)
(11)		CDD18.3	36	0	0	0	0	2	6	14	13	1	0	0	0	(11)
(12)		CDH23.3	549	0	0	0	2	63	78	181	168	56	1	0	0	(12)
(13)		CDH26.7	84	0	0	0	0	9	10	24	29	11	0	0	0	(13)
(14)	Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(14)
(15)	Precipitation	PrecAvg	431	22	11	17	25	49	82	82	56	35	19	20	18	(15)
(16)		PrecMax	538	93	23	38	49	79	134	161	118	79	34	70	33	(16)
(17)		PrecMin	321	0	0	3	1	7	32	38	22	4	2	3	1	(17)
(18)		PrecStd	71	22	7	10	13	19	29	31	22	19	10	17	10	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.2	7.0	11.3	22.6	27.8	28.2	29.2	29.6	28.4	21.6	12.4	5.6	(19)
(20)			MCWB	2.6	2.9	6.1	10.9	13.2	16.9	19.8	17.3	15.8	12.1	7.1	2.0	(20)
(21)		2%	DB	5.0	4.1	7.8	18.0	25.1	25.6	26.9	27.0	24.7	16.6	8.0	3.7	(21)
(22)			MCWB	1.9	1.2	3.6	8.2	12.2	15.7	18.4	17.8	14.6	9.2	4.5	0.8	(22)
(23)		5%	DB	3.7	2.5	5.9	14.6	22.5	23.6	25.4	25.1	22.2	13.9	5.8	1.8	(23)
(24)			MCWB	1.3	0.3	2.6	6.6	11.2	14.8	17.6	17.1	13.3	7.7	2.7	-0.5	(24)
(25)		10%	DB	2.0	0.7	4.0	12.0	20.2	21.8	23.7	23.4	19.3	11.5	3.6	-0.4	(25)
(26)			MCWB	0.1	-1.1	1.2	5.3	10.2	14.0	16.7	16.2	12.3	6.5	1.6	-2.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.1	3.2	6.3	10.9	14.3	18.6	21.2	20.5	16.8	12.5	7.4	2.5	(27)
(28)			MCDWB	5.7	6.3	10.6	21.9	23.1	25.3	26.8	26.2	25.1	20.0	11.9	5.1	(28)
(29)		2%	WB	2.2	1.7	4.1	8.6	13.2	16.9	19.7	19.1	15.2	10.1	4.7	1.0	(29)
(30)			MCDWB	4.6	3.8	7.2	16.9	22.2	23.1	25.3	24.9	23.0	15.3	7.5	3.3	(30)
(31)		5%	WB	1.4	0.3	2.7	7.1	12.1	15.8	18.5	17.9	14.0	8.3	3.0	-0.5	(31)
(32)			MCDWB	3.5	2.2	5.6	13.8	20.9	21.8	23.8	23.4	21.0	12.9	5.1	1.8	(32)
(33)		10%	WB	0.1	-1.0	1.4	5.8	11.0	14.9	17.4	16.7	12.8	7.0	1.7	-2.4	(33)
(34)			MCDWB	2.2	1.1	3.7	11.1	18.4	20.3	22.4	22.0	18.5	10.7	3.5	-0.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.6	8.9	8.8	10.3	12.4	10.9	11.5	12.0	11.8	9.6	7.6	8.0	(35)
(36)			MCDBR	7.4	9.2	9.9	15.3	16.0	13.8	14.3	14.7	16.9	13.4	9.7	9.7	(36)
(37)			MCWBR	5.8	7.3	6.6	7.8	6.7	6.2	7.3	6.9	7.9	7.6	6.6	8.0	(37)
(38)		5% WB	MCDBR	7.4	8.8	9.6	14.3	14.5	12.0	13.1	13.4	15.9	12.9	8.8	9.6	(38)
(39)			MCWBR	6.1	7.2	6.6	7.7	6.2	6.0	7.3	6.9	7.7	7.5	6.4	8.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.254	0.270	0.308	0.324	0.339	0.351	0.352	0.359	0.305	0.280	0.251	0.236	(40)	
(41)		taud	2.294	2.325	2.301	2.392	2.399	2.377	2.420	2.350	2.507	2.549	2.453	2.310	(41)	
(42)		Ebn at Noon	725	839	873	908	908	897	888	859	877	831	745	680	(42)	
(43)		Edh at Noon	56	77	99	104	109	113	106	108	80	62	48	45	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.91	1.94	3.46	4.66	5.70	5.98	6.08	4.99	3.44	1.95	0.92	0.66	(44)	
(45)		RadStd	0.10	0.14	0.26	0.41	0.45	0.40	0.31	0.35	0.35	0.25	0.10	0.06	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (59 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page